

### 6.1.1 Electromagnetism

| Learning outcomes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Additional guidance                                                                                                                                          |
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| <p><b>(a)</b> <i>Describe and explain:</i></p> <p><b>(i)</b> the action of a transformer: magnetic flux from a coil; induced e.m.f = rate of change of flux linkage</p> <p><b>(ii)</b> the action of a dynamo: change of flux linked produced by relative motion of flux and conductor</p> <p><b>(iii)</b> electromagnetic forces; qualitatively as arising from tendency of flux lines to contract or interaction of induced poles; quantitative calculation limited to force on a straight current-carrying wire in a uniform field</p> <p><b>(iv)</b> simple linked electric and magnetic circuits: flux produced by current turns, need for large conductance and permeance and the effect of increasing the dimensions of an electromagnetic machine; qualitative effect of iron and air gap.</p> | HSW2                                                                                                                                                         |
| <p><b>(b)</b> <i>Make appropriate use of:</i></p> <p><b>(i)</b> the terms: <b>B</b>-field, magnetic field, flux, flux linkage, induced e.m.f, eddy currents</p> <p><i>by sketching and interpreting:</i></p> <p><b>(ii)</b> graphs of variations of currents, flux and induced e.m.f</p> <p><b>(iii)</b> diagrams of lines of flux in magnetic circuits; continuity of lines of flux.</p>                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                              |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | permeance is understood as a magnetic equivalent to electrical conductance and with analogous dependence on the dimensions and nature of the magnetic medium |

**(c)** *Make calculations and estimates involving:*

**(i)**  $\phi = BA, \mathcal{E} = -\frac{d(\phi N)}{dt}$

In the context of Faraday's and Lenz's laws  
HSW1, 2, 8

**(ii)**  $F = ILB$

Calculations restricted to current perpendicular to  
uniform magnetic field

**(iii)**  $\frac{V_1}{V_2} = \frac{N_1}{N_2}$  for an ideal transformer

i.e. no energy losses for an ideal transformer

Learners will also be expected to recall the equation  
relating  $V$  and  $N$

**(iv)**  $\frac{I_2}{I_1} = \frac{N_1}{N_2}$  for an ideal transformer

Learners will also be expected to recall the equation  
relating  $I$  and  $N$   
HSW12

**(d)** *Demonstrate and apply knowledge and  
understanding of the following practical activities  
(HSW4):*

- (i)** observing induced e.m.fs produced under  
varying conditions such as dropping a  
magnet through a coil attached to a data  
logger or oscilloscope

links to **6.1.1a(ii)**, **6.1.1c(i)**

- (ii)** determining the uniform magnetic flux  
density between the poles of a magnet  
using a rigid current carrier and digital  
balance.

links to **6.1.1a(iii)**, **c(ii)**

- (iii)** investigate transformers.

links to **6.1.1a(i)**, **c(iii)(iv)**